

GRADE 9 MATHEMATICS · MEASUREMENT & GEOMETRY

Similarity & Scale

THE BIG IDEA

**Similar figures are the same
— angles match exactly, side
factor.**

TWO POLYGONS ARE SIMILAR ONLY IF

1. All corresponding angles are **equal**, AND

RY

**the shape at a different size
sides scale by the same**

ND

2. All corresponding sides are **proportion**

Triangles shortcut — **AA**: two pairs of equal angl

WORKED EXAMPLE

$$\triangle ABC \sim \triangle DEF \quad AB=6, BC=9, AC=12, DE=4$$

$$\text{Scale factor} = DE \div AB = 4/6 = \mathbf{2/3}$$

$$EF = 9 \times 2/3 = \mathbf{6}$$

$$DF = 12 \times 2/3 = \mathbf{8}$$

⚠ WATCH OUT FOR

MISTAKE

“Similar means congruent”

MISTAKE

es is enough (the third must match too).

=4

FIX

Similar means proportional — **not** equal in size

FIX

Setting up the ratio backwards

MISTAKE

Measuring shadows at different times



SELF-CHECK

If $\triangle PQR \sim \triangle XYZ$ with scale factor $3/2$, and PO

FIX

Sun angle must stay constant — measure simultaneously

$Q = 4$ cm, what is XY ?